

Technical Admin Manual: Medical Calculator Suite

Project Version: 1.0

Release Date: May 2026

System Requirements: Windows 10/11, Python 3.10.1+

1. System Architecture Overview

This application follows a modular architecture to separate the User Interface (Tkinter) from the Data Processing (JSON/Logic).

- **Entry Point:** `Main.py` (Manages window initialization, fixed scaling, and navigation).
 - **Front_end/:** Contains UI logic scripts for Login, Account Creation, and all Calculator modules.
 - **Back_end/:** Contains `database.py` (User Auth & History) and formula logic.
 - **Assets/:** Contains all `.png` image assets for buttons and backgrounds.
-

2. Environment Setup & Installation

To recreate the development environment or install the app on a new machine:

1. **Install Python:** Download Python 3.10.1 or higher from python.org.
 2. **Automated Setup (ONLY DO THIS IF THE APP IS NOT RUNNING OR YOU WANT TO CHANGE/CHECK THE CODE) :** Double-click `setup.bat` in the root directory. This script will automatically run `pip install -r requirements.txt`.
 3. **Directory Integrity:** Ensure the `assets/` folder remains in the root directory. Moving this folder will cause the application to crash on startup.
-

3. Data Management

The system utilizes local JSON files for data persistence.

- **User Accounts (`users.json`):** Stores encrypted/hashed credentials. Deleting this file will reset all user data.
- **Calculation History (`history.json`):** Stores timestamped logs of all calculations.

- **Backup Procedure:** It is recommended to back up these two files weekly. In case of corruption, replacing these files with the latest backup will restore all system data.
-

4. Maintenance & Troubleshooting

Issue	Potential Cause	Resolution
App won't start	Missing <code>assets/</code> folder.	Verify <code>assets/</code> is in the same folder as <code>Main.py</code> .
Login fails for everyone	<code>users.json</code> is corrupted.	Restore from backup or delete to allow fresh registrations.
UI looks "broken"	Windows Display Scaling > 100%.	Set OS Display Scaling to 100% or check <code>root.resizable</code> logic.
Calculation Error	Logic/Formula bug.	Verify <code>Back_end/</code> logic against the Validation Matrix below.

5. Deployment (Creating the EXE)

To package a new version for the client using **Auto Py to Exe**:

1. Set **Script Location** to `Main.py`.
 2. Select **One Directory** (for speed) or **One File** (for portability).
 3. Set **Console Window** to "Window Based" (to hide the black terminal).
 4. **Crucial:** In "Additional Files," add the folders: `Front_end/`, `Back_end/`, and `assets/`.
 5. Click **Convert .py to .exe**.
-

6. Software Bill of Materials (SBOM)

This application relies on the following components:

- **Python Standard Library:** `tkinter`, `json`, `pathlib`, `sqlite3`.
 - **Pillow (v8.4.0):** Used for advanced image processing and rendering.
 - **Custom Assets:** Proprietary `.png` designs for the "Medical Red" themed UI.
-

7. Security & Compliance

- **Data Locality:** All patient-related calculations are stored locally on the machine. No data is transmitted to external servers (HIPAA/GDPR compliant design).
 - **Access Control:** Access to history logs is restricted to the specific logged-in user.
-